Volumes of revolution and average values

1. Disks and washers

A *solid of revolution* is the set swept out by a plane region during one full turn about a line in its plane. Let t measure position along this axis from a to b. If the perpendicular section at t has continuous area $S(t)$, the slicing formula gives $V = \int_a^b S(t)\,dt$. In the original plane, take a segment perpendicular to the axis. A segment whose distances from the axis run from r to R sweeps out the ring between two circles. Such a section is called a *washer*. Subtracting the inner disk area from the outer disk area gives

$$S(t) = \pi R(t)^2 - \pi r(t)^2, \qquad \boxed{V = \pi \int_a^b \big(R(t)^2 - r(t)^2\big)\,dt.}$$

Here each slice is a single segment and $0 \le r(t) \le R(t)$. Continuous radii suffice, and the contributions from adjacent intervals add. When the segment meets the axis, $r = 0$ and its section is a disk.

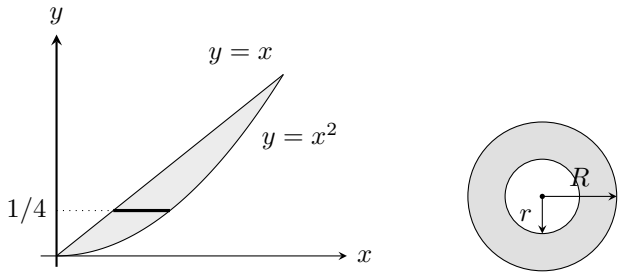
Example

A right circular cone has height h and base radius B. Measure t along its axis from the tip. Similar triangles give $R(t)/t = B/h$ for $0 < t \le h$, and $R(0) = 0$. Thus

$$V = \pi \int_0^h (Bt/h)^2\,dt = \frac{\pi B^2 h}{3}.$$

2. A washer about the y-axis

Rotate $D = \{(x, y) : 0 \le x \le 1,\ x^2 \le y \le x\}$ about the y-axis. At a fixed height y between 0 and 1, the inequalities give $y \le x \le \sqrt{y}$. A horizontal segment therefore starts at radius y and ends at radius $\sqrt{y}$.



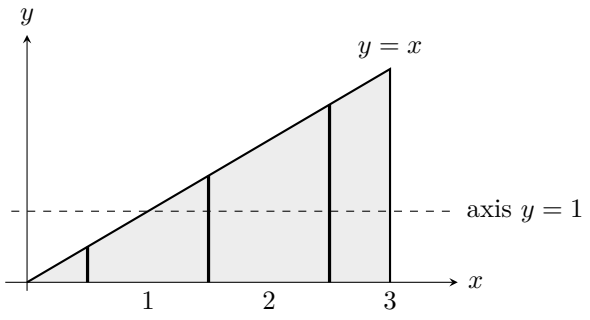
At $y = 1/4$ the two radii are $1/4$ and $1/2$, so the section has area $3\pi/16$. For the whole solid,

$$V = \pi \int_0^1 \big((\sqrt{y})^2 - y^2\big)\,dy = \pi \left[\frac{y^2}{2} - \frac{y^3}{3}\right]_0^1 = \frac{\pi}{6}.$$

3. An axis passing through the region

Example

Rotate the triangle $0 \le x \le 3$, $0 \le y \le x$, about $y = 1$. At each fixed x, the bottom endpoint of the vertical segment is one unit from the axis. The top endpoint is $|x - 1|$ units away.



At $x = 1/2$ the section has radii $1/2$ and 1. At $x = 3/2$ the segment crosses the axis, and the bottom endpoint sweeps the outer circle of radius 1. At $x = 5/2$ the top endpoint sweeps the outer circle of radius $3/2$. When both sides of the segment reach a given radius, their turns cover the same circle.

Interval	Section	Outer radius	Inner radius
$0 \le x \le 1$	Washer (hole closes at 1)	1	$1 - x$
$1 \le x \le 2$	Disk	1	0
$2 \le x \le 3$	Disk	$x - 1$	0

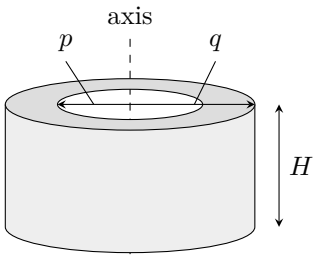
The first change occurs where the upper endpoint reaches the axis. The second occurs where its distance from the axis equals that of the lower endpoint, so $x - 1 = 1$. Thus

$$\begin{aligned} V &= \pi \int_0^1 \bigl(1 - (1 - x)^2\bigr)\, dx + \pi \int_1^2 1\, dx + \pi \int_2^3 (x - 1)^2\, dx \\ &= \pi \left(\frac{2}{3} + 1 + \frac{7}{3} \right) = 4\pi. \end{aligned}$$

The three marked sections have areas $3\pi/4,\ \pi,\ 9\pi/4$, respectively.

4. Cylindrical shells

Rotating a rectangle of height $H > 0$, with vertical sides at $x = p$ and $x = q$, about the y-axis produces a *cylindrical shell* of inner radius p, outer radius q, and height H. Here $0 \le p < q$.



Factoring the difference of the two cylinder volumes gives

$$\pi(q^2 - p^2)H = 2\pi \underbrace{\frac{p+q}{2}}_{\text{mean radius}} H \underbrace{(q-p)}_{\text{thickness}}.$$

This equality is exact for a rectangular strip of any thickness.

Now rotate a region $a \le x \le b$, $g(x) \le y \le f(x)$ about the y-axis, with $0 \le a < b$ and continuous f, g. Its vertical strip has height $h(x) = f(x) - g(x)$. Partition $[a, b]$ into intervals of widths Δx_i with midpoints m_i. Rotating the rectangles of height $h(m_i)$ gives shells of mean radius m_i. As the largest width tends to zero,

$$\sum_i 2\pi m_i \, h(m_i) \, \Delta x_i \longrightarrow \boxed{V = 2\pi \int_a^b x \, h(x) \, dx}.$$

Justification. First suppose $g = 0$, so $f \ge 0$. On each interval use the minimum and maximum of f as the heights of two rectangular strips. Their shells enclose the corresponding part of the solid. The total difference in volume is at most the largest variation in height on any subinterval times the sum of the annular base areas, $\sum_i \pi(x_i^2 - x_{i-1}^2) = \pi(b^2 - a^2)$. Since f is uniformly continuous on $[a, b]$, the largest difference between a maximum and minimum tends to zero with the largest width. The midpoint sum lies between the same bounds. For general $g \le f$, take a horizontal baseline below both graphs and subtract the two volumes above it.

For rotation about $x = c$ the radius is $|x - c|$. When the region lies on one side of the axis, different x-values give distinct radii. For a region on both sides, strips at equal distances can sweep out the same volume. Account for their overlap before adding shells.

5. Washers and shells for the same solid

Consider again the region between $y = x^2$ and $y = x$ for $0 \le x \le 1$, rotated about the y-axis. A vertical strip at $x = 1/2$ has height $1/2 - 1/4 = 1/4$ and lies $1/2$ unit from the axis. At a general x, the shell radius is x and its height is $x - x^2$. Thus

$$V = 2\pi \int_0^1 x(x - x^2)\,dx = 2\pi \left[\frac{x^3}{3} - \frac{x^4}{4}\right]_0^1 = \frac{\pi}{6}.$$

	Horizontal strip	Vertical strip
Rotation about y-axis	Washer	Cylindrical shell
Dimensions	$R = \sqrt{y}, \quad r = y$	Radius x, height $x - x^2$
Thickness	dy	dx

The geometry gives the identity

$$\pi \int_0^1 (y - y^2)\,dy = 2\pi \int_0^1 x(x - x^2)\,dx.$$

For washers, we solved for the two x-coordinates at each height y. For shells, the difference between the given functions supplied the height directly.

Example

Rotate the rectangle $1 \le x \le 2$, $0 \le y \le 3$ about the y-axis. A horizontal strip makes a washer of radii 2 and 1. A vertical strip makes a shell of radius x and height 3. Both give

$$\pi \int_0^3 (2^2 - 1^2)\,dy = 2\pi \int_1^2 3x\,dx = 9\pi.$$

6. Rotation about a shifted horizontal axis

Example

Rotate the region between $y = x^2$ and $y = x$ on $[0, 1]$ about the horizontal line $y = -1$. A horizontal strip at y now has radius $1 + y$ and thickness dy. Its length parallel to the axis is $\sqrt{y} - y$, which gives the shell height. A vertical strip gives a washer with outer radius $1 + x$ and inner radius $1 + x^2$. The two setups give

$$\begin{aligned} 2\pi \int_0^1 (1 + y)(\sqrt{y} - y)\,dy &= \pi \int_0^1 \left((1 + x)^2 - (1 + x^2)^2\right) dx \\ &= \pi \int_0^1 (2x - x^2 - x^4)\,dx = \frac{7\pi}{15}. \end{aligned}$$

7. Average value of a function

Let f be Riemann integrable on $[a, b]$, where $a < b$. Its *average value* on the interval is

$$\boxed{f_{\text{avg}} = \frac{1}{b-a}\int_a^b f(x)\,dx.}$$

To see why the interval length appears, divide $[a, b]$ into n equal pieces of width $\Delta x = (b-a)/n$. Choose a point ξ_i in each piece. The ordinary average of the sampled values is

$$\frac{f(\xi_1) + \cdots + f(\xi_n)}{n} = \frac{1}{b-a}\sum_{i=1}^{n} f(\xi_i)\,\Delta x.$$

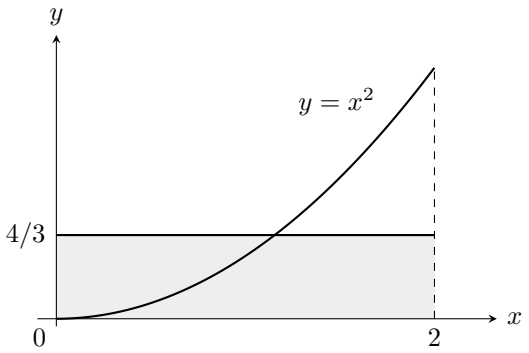
The sum on the right is a Riemann sum. As n increases, the average of the sampled values tends to f_{avg}.

The constant function with height f_{avg} has the same integral on $[a, b]$ as f. For $f \geq 0$, this means that the rectangle with base $b-a$ and height f_{avg} has the same area as the region under the graph. When f takes negative values, the integral counts the contributions below the axis with a minus sign.

Example

For $f(x) = x^2$ on $[0, 2]$, the area under the graph is $\int_0^2 x^2\,dx = 8/3$. The interval has length 2, so

$$f_{\text{avg}} = \frac{1}{2}\left[\frac{x^3}{3}\right]_0^2 = \frac{4}{3}.$$



The shaded rectangle has area $2 \cdot (4/3) = 8/3$.

For the cone in Section 1, the same formula gives the average cross-sectional area. The integral of $S(t)$ from 0 to h is the volume of the cone, so

$$S_{\text{avg}} = \frac{1}{h}\int_0^h S(t)\,dt = \frac{1}{h}\cdot\frac{\pi B^2 h}{3} = \frac{\pi B^2}{3}.$$